Year 11 Higher

REVISION PACK

Number 8

Name: _____

Class: _____

Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

Worked example: prime factors in index form

Write 252 as a product of its prime factors.
Give your answer in index form.

Step 1	Divide by 2	$252 = 2 \times 126$
Step 2	Divide by 2	$126 = 2 \times 63$
Step 3	Divide by 3	$63 = 3 \times 21$
Step 4	Divide by 3	$21 = 3 \times 7$
Step 5	Answer	$2^2 \times 3^2 \times 7$

Practice: prime factors in index form

1.1 Write 120 as a product of its prime factors. Give your answer in index form.

1.2 Write 504 as a product of its prime factors. Give your answer in index form.

1.3 Write 600 as a product of its prime factors. Give your answer in index form.

1.4 Write 280 as a product of its prime factors. Give your answer in index form.

1.5 Write 450 as a product of its prime factors. Give your answer in index form.

1.6 Write 792 as a product of its prime factors. Give your answer in index form.

Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

Worked example: HCF and LCM

Find

(a) the highest common factor (HCF) of 60 and 72 (b) the lowest common multiple (LCM) of 60 and 72

(a)	**Step 1**	Prime factors of 60	$60 = 2^2 \times 3 \times 5$
	Step 2	Prime factors of 72	$72 = 2^3 \times 3^2$
	Step 3	HCF: primes in both	$2^2 \times 3 = 12$
(b)	**Step 1**	LCM: highest powers	$2^3 \times 3^2 \times 5 = 360$

Practice: HCF and LCM

1.7 Find

(a) the highest common factor (HCF) of 90 and 150 (b) the lowest common multiple (LCM) of 90 and 150

1.8 Find

(a) the highest common factor (HCF) of 48 and 180 (b) the lowest common multiple (LCM) of 48 and 180

1.9 Find

(a) the highest common factor (HCF) of 72 and 108 (b) the lowest common multiple (LCM) of 72 and 108

1.10 Find

(a) the highest common factor (HCF) of 56 and 84 (b) the lowest common multiple (LCM) of 56 and 84

1.11 Find

(a) the highest common factor (HCF) of 126 and 168 (b) the lowest common multiple (LCM) of 126 and 168

1.12 Find

(a) the highest common factor (HCF) of 60 and 96 (b) the lowest common multiple (LCM) of 60 and 96

Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

Worked example: next time together

Bus A leaves the bus station every 15 minutes. Bus B leaves the bus station every 20 minutes. Both buses leave the bus station at 9:00 am. When will both buses next leave the bus station at the same time?

Step 1	LCM of 15 and 20	60 minutes
Step 2	Add to 9:00 am	$9:00 + 60$ minutes
Step 3	Answer	$10:00$ am

Practice: next time together

1.13 Bus A leaves the bus station every 16 minutes. Bus B leaves the bus station every 24 minutes. Both buses leave the bus station at 7:00 am. When will both buses next leave the bus station at the same time?

1.14 Train A leaves the station every 10 minutes. Train B leaves the station every 25 minutes. Both trains leave the station at 8:30 am. When will both trains next leave the station at the same time?

1.15 Bus A leaves the bus station every 14 minutes. Bus B leaves the bus station every 21 minutes. Both buses leave the bus station at 6:00 am. When will both buses next leave the bus station at the same time?

1.16 Train A leaves the station every 20 minutes. Train B leaves the station every 30 minutes. Both trains leave the station at 9:15 am. When will both trains next leave the station at the same time?

1.17 Bus A leaves the bus station every 18 minutes. Bus B leaves the bus station every 24 minutes. Both buses leave the bus station at 7:30 am. When will both buses next leave the bus station at the same time?

1.18 Bus A leaves the bus station every 12 minutes. Bus B leaves the bus station every 15 minutes. Both buses leave the bus station at 10:00 am. When will both buses next leave the bus station at the same time?

Number 8 · Standard form (Lesson 1.1)

Worked example: calculating in standard form

Work out. Give your answers in standard form.

(a) $(2.5 \times 10^3) \times (6 \times 10^{-5})$ (b) $(3 \times 10^4) + (8.2 \times 10^3)$

(a)	**Step 1**	Multiply the numbers	$2.5 \times 6 = 15$
	Step 2	Add the powers	$10^{3+(-5)} = 10^{-2}$
	Step 3	15 = 1.5 × 10: power +1	1.5×10^{-1}
(b)	**Step 1**	Change to ordinary numbers	$30\,000 + 8200 = 38\,200$
	Step 2	Answer	3.82×10^4

Practice: calculating in standard form

2.1 Work out. Give your answers in standard form.

(a) $(4 \times 10^3) \times (2.5 \times 10^{-6})$ (b) $(5 \times 10^4) + (3.2 \times 10^3)$

2.2 Work out. Give your answers in standard form.

(a) $(1.6 \times 10^5) \times (5 \times 10^{-2})$ (b) $(7 \times 10^5) + (6.5 \times 10^4)$

2.3 Work out. Give your answers in standard form.

(a) $(2.4 \times 10^{-3}) \times (3 \times 10^8)$ (b) $(2 \times 10^6) + (8.5 \times 10^5)$

2.4 Work out. Give your answers in standard form.

(a) $(6 \times 10^4) \times (1.5 \times 10^{-9})$ (b) $(4.1 \times 10^3) + (9 \times 10^2)$

2.5 Work out. Give your answers in standard form.

(a) $(3.5 \times 10^2) \times (4 \times 10^{-5})$ (b) $(8 \times 10^4) + (2.4 \times 10^3)$

2.6 Work out. Give your answers in standard form.

(a) $(2.5 \times 10^{-4}) \times (8 \times 10^7)$ (b) $(1.2 \times 10^5) + (6 \times 10^4)$

Number 8 · Standard form (Lesson 1.1)

Worked example: writing numbers in standard form

This question is about standard form.

(a) Write 0.000507 in standard form. (b) Write 2.4×10^{-4} as an ordinary number.

(a)	Step 1	Move point to after the 5	5.07
	Step 2	Point moved 4 right: power −4	5.07×10^{-4}
(b)	Step 1	Power −4: divide by 10 000	$2.4 \div 10\,000 = 0.00024$

Practice: writing numbers in standard form

2.7 This question is about standard form.

(a) Write 0.00072 in standard form.

(b) Write 3.6×10^{-4} as an ordinary number.

2.8 This question is about standard form.

(a) Write 0.0000415 in standard form.

(b) Write 5.2×10^{-3} as an ordinary number.

2.9 This question is about standard form.

(a) Write 0.00908 in standard form.

(b) Write 1.75×10^{-5} as an ordinary number.

2.10 This question is about standard form.

(a) Write 0.000063 in standard form.

(b) Write 8.04×10^{-2} as an ordinary number.

2.11 This question is about standard form.

(a) Write 0.0123 in standard form.

(b) Write 4.9×10^{-6} as an ordinary number.

2.12 This question is about standard form.

(a) Write 0.000000208 in standard form.

(b) Write 6.3×10^{-3} as an ordinary number.

Number 8 · Ratio (Lesson 1.4)

Worked example: sharing in a ratio

Share 240 KD in the ratio 1 : 3 : 4

Step 1	Add the parts	$1 + 3 + 4 = 8$
Step 2	One part	$240 \div 8 = 30$
Step 3	1 part	$1 \times 30 = 30$
Step 4	3 parts	$3 \times 30 = 90$
Step 5	4 parts	$4 \times 30 = 120$
Step 6	Answer	$30\ \text{KD},\ 90\ \text{KD},\ 120\ \text{KD}$

Practice: sharing in a ratio

3.1 Share 300 KD in the ratio 2 : 3 : 5

3.2 Share 420 KD in the ratio 1 : 2 : 4

3.3 Share 540 KD in the ratio 2 : 3 : 4

3.4 Share 600 KD in the ratio 3 : 4 : 5

3.5 Share 330 KD in the ratio 1 : 4 : 6

3.6 Share 720 KD in the ratio 1 : 3 : 5

Number 8 · Ratio (Lesson 1.4)

Worked example: ratio and difference

The ratio of boys to girls in a club is 3 : 5. There are 14 more girls than boys. How many boys and girls are there altogether?

Step 1	Difference in parts	$5 - 3 = 2$
Step 2	One part	$14 \div 2 = 7$
Step 3	Total parts	$3 + 5 = 8$
Step 4	Total parts × one part	$8 \times 7 = 56$

Practice: ratio and difference

3.7 The ratio of boys to girls in a club is 2 : 5. There are 18 more girls than boys. How many boys and girls are there altogether?

3.8 The ratio of cats to dogs in a shelter is 3 : 7. There are 20 more dogs than cats. How many cats and dogs are there altogether?

3.9 The ratio of red cars to blue cars in a car park is 4 : 7. There are 15 more blue cars than red cars. How many red cars and blue cars are there altogether?

3.10 The ratio of adults to children in a cinema is 5 : 8. There are 12 more children than adults. How many adults and children are there altogether?

3.11 The ratio of teachers to students in a school trip is 1 : 4. There are 27 more students than teachers. How many teachers and students are there altogether?

3.12 The ratio of boys to girls in a class is 3 : 4. There are 9 more girls than boys. How many boys and girls are there altogether?